

UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0264: Generative Artificial Intelligence

Master Past Exam Revision Guide & Exhaustive Solutions

Compiling Examinations and Core Concepts for Cohorts 2025–2026

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Master Course Syllabus & Core Roadmap

This master revision guide compiles the examination papers and core theoretical foundations for UCL COMP0264 (Generative Artificial Intelligence):

1. **2025–2026 Mock Examination Paper (Variant A):** Encoder/Decoder VAEs, Posterior Collapse, Transformers, LLMs, and Diffusion Models.
2. **2025–2026 Mock Examination Paper (Variant B):** Generator/Discriminator VAEs vs GANs, Mode Collapse, and Scaling Laws.

The course covers four primary pillars of modern deep generative modeling:

Topic Area	Core Concepts and Key Formulations
1. VAEs & Latent Models	Encoder $q_\phi(z x)$, Decoder $p_\theta(x z)$, Reparameterization Trick, ELBO loss, Posterior Collapse, β -VAE.
2. GANs & Adversarial Models	Generator G_θ , Discriminator D_ϕ , Minimax game $V(G, D)$, JS-divergence, Mode Collapse, WGAN-GP.
3. Transformers & LLMs	Self-Attention, Scaled Dot-Product $\frac{QK^T}{\sqrt{d_k}}$, Positional Encoding (Sinusoidal, RoPE), Masking, Emergent ICL, RLHF/DPO.
4. Diffusion Models	Forward Process $q(x_t x_0)$, Reverse Process $p_\theta(x_{t-1} x_t)$, Reparameterization $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$, Denoising Loss.

PART 1: COMPREHENSIVE MOCK EXAMINATION PAPERS

PAPER 1: COMP0264 Mock Examination 2025–2026

Duration: 3 Hours — Total Marks: 100

Question 1: Variational Autoencoders & GANs [25 Marks]

- (a) Describe the architecture of the encoder and decoder of a Variational Autoencoder (VAE). Discuss the training procedure including the loss function (ELBO) used for optimization. [12.5 Marks]
- (b) Explain why training VAEs can be unstable and discuss the causes of the potential problem of posterior collapse (i.e. the situation in which the encoder's output becomes almost identical to the prior, regardless of the input). Compare this to mode collapse in GANs and discuss possible solutions. [12.5 Marks]

Question 2: Transformer Architectures [25 Marks]

- (a) Explain the self-attention mechanism in transformer architectures and why it is effective compared to recurrent neural networks. [5 Marks]
- (b) Why is positional encoding necessary in transformers? Describe one method for incorporating positional information (e.g. sinusoidal or RoPE). [10 Marks]
- (c) Discuss why we need scaled self-attention. Provide a mathematical justification by deriving the variance scaling factor $\sqrt{d_k}$. [10 Marks]

Question 3: Large Language Models (LLMs) [25 Marks]

- (a) Despite being trained with a next-token prediction objective, LLMs exhibit emergent behaviors such as in-context learning (ICL). Provide a critical explanation, grounded in probabilistic modeling, of why such behavior arises. [5 Marks]
- (b) In transformer-based LLMs, the conditional distribution $p(x_t \mid x_{<t})$ is parameterized using masked self-attention. Explain how masking ensures the autoregressive property is preserved during training. [5 Marks]
- (c) Suppose an LLM assigns high probability to fluent but factually incorrect sequences. Explain why maximum likelihood training does not guarantee factual correctness, and discuss one potential solution (e.g. RLHF/DPO or RAG). [7.5 Marks]
- (d) Explain how fine-tuning can lead to catastrophic forgetting from a probabilistic perspective. Use the Kullback-Leibler (KL) divergence to formalize the intuition. [7.5 Marks]

Question 4: Diffusion Models [25 Marks]

- (a) Describe the forward and reverse processes in a diffusion model. What is the goal of each process? Formulate the underlying mathematical model for both processes and explain how the reverse process is derived from the forward process. [12.5 Marks]
- (b) How is the reverse decoder in a diffusion model trained? What computational challenges arise in doing so, and what practical approach (ϵ -prediction MSE objective) is used instead? [12.5 Marks]

PART 2: EXHAUSTIVE MASTER SOLUTIONS GUIDE

Solution 1: Variational Autoencoders & GANs

Part (a): VAE Architecture, Reparameterization Trick & ELBO Derivation

1. **Encoder and Decoder Architecture:** A VAE consists of two neural networks:

- **Encoder** $q_\phi(z | x)$: Maps high-dimensional input $x \in \mathbb{R}^D$ to the parameters of a continuous latent distribution (typically a Gaussian $\mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$). Outputs mean $\mu_\phi(x)$ and log-variance $\log \sigma_\phi^2(x)$.
 - **Decoder** $p_\theta(x | z)$: Maps latent code $z \in \mathbb{R}^d$ ($d \ll D$) back to data space, parameterizing distribution $p_\theta(x | z)$ (e.g. Bernoulli for binary data or Gaussian $\mathcal{N}(\mu_\theta(z), \sigma^2 I)$ for continuous data).
2. **The Reparameterization Trick:** Direct sampling $z \sim \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x))$ is non-differentiable with respect to encoder parameters ϕ . The reparameterization trick isolates stochasticity by sampling an independent noise vector $\epsilon \sim \mathcal{N}(0, I)$ and expressing z deterministically as:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$$

This allows backpropagation of gradients ∇_ϕ via standard chain rule through μ_ϕ and σ_ϕ .

3. **Mathematical Derivation of the Evidence Lower Bound (ELBO):** We wish to maximize true data log-likelihood $\log p_\theta(x) = \log \int p_\theta(x, z) dz$. Introducing variational posterior $q_\phi(z | x)$:

$$\log p_\theta(x) = \log \int q_\phi(z | x) \frac{p_\theta(x, z)}{q_\phi(z | x)} dz = \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z | x)} \right]$$

Applying **Jensen's Inequality** ($\log \mathbb{E}[X] \geq \mathbb{E}[\log X]$ since $\log$ is concave):

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z | x)} \right] \equiv \mathcal{L}_{\text{ELBO}}(\theta, \phi)$$

Expanding $\mathcal{L}_{\text{ELBO}}$:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}}(\theta, \phi) &= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x | z)p(z)}{q_\phi(z | x)} \right] \\ &= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z | x)}{p(z)} \right] \\ &= \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)]}_{\text{Reconstruction Loss}} - \underbrace{KL(q_\phi(z | x) \parallel p(z))}_{\text{KL Regularization Term}} \end{aligned}$$

For Gaussian prior $p(z) = \mathcal{N}(0, I)$ and encoder $q_\phi(z|x) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$, the KL divergence integrates analytically:

$$KL(q_\phi(z | x) \parallel p(z)) = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

Part (b): Posterior Collapse in VAEs vs Mode Collapse in GANs

1. **Posterior Collapse in VAEs:** Posterior collapse occurs when the variational posterior matches the prior independent of input: $q_\phi(z | x) \approx p(z) = \mathcal{N}(0, I)$, causing $KL(q_\phi(z | x) \parallel p(z)) \rightarrow 0$. The encoder completely ignores input x , and the decoder ignores latent z .

- **Cause:** Occurs when using an overly powerful autoregressive decoder (e.g. PixelCNN or Transformer). A strong decoder can model $p(x)$ directly without using z . Since $KL \rightarrow 0$ maximizes the ELBO objective, the optimizer chooses the degenerate path of setting $q(z|x) = p(z)$.
- **Solutions:**

- (a) **β -VAE Annealing:** Multiply KL term by weight β . Gradually increase β from 0 to 1 during training (β schedule), forcing the model to learn meaningful z early when $\beta = 0$.
 - (b) **Free Bits / KL Minimum Margin:** Enforce a lower bound threshold λ on individual latent dimensions: $\max(\lambda, KL(q_\phi(z_i|x) \parallel p(z_i)))$.
 - (c) **VQ-VAE (Vector Quantized VAE):** Replace continuous Gaussian latent space with discrete codebook vectors, preventing KL collapse.
2. **Mode Collapse in GANs:** In GANs ($\min_G \max_D V(G, D) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))]$), mode collapse occurs when generator G learns to produce samples from only a single mode or narrow subset of data modes (e.g. generating only 1 digit out of 10 in MNIST).
- **Cause:** GAN objective optimizes Jensen-Shannon divergence $JS(p_{data} \parallel p_g)$ when D is optimal. JS divergence is zero-forcing for missing modes, allowing G to over-optimize on a single real-looking mode that fools D .
 - **Solutions:** Wasserstein GAN with Gradient Penalty (WGAN-GP) using Earth Mover's distance, spectral normalization of D , and unrolled GANs.

Solution 2: Transformer Architectures

Part (a): Self-Attention Mechanism

Self-attention maps input sequence $X \in \mathbb{R}^{N \times d_{\text{model}}}$ to Queries (Q), Keys (K), and Values (V):

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

where $W_Q, W_K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $W_V \in \mathbb{R}^{d_{\text{model}} \times d_v}$. Attention output matrix:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Effectiveness: 1. ****Dynamic Content Routing:**** Computes pairwise similarity between all token pairs directly ($O(1)$ path length), eliminating vanishing gradients over long sequences. 2. ****Parallel Training:**** Unlike RNNs ($h_t = f(h_{t-1}, x_t)$ which require sequential processing), matrix product QK^T processes all N tokens simultaneously on GPUs.

Part (b): Positional Encoding Necessity & Formulations

Self-attention is ****permutation equivariant****: permuting input rows PX yields permuted outputs $P\text{Attention}(X)$. Without positional information, a transformer treats a sentence as an unordered bag of words.

Sinusoidal Positional Encoding (Vaswani et al. 2017):

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \quad PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right)$$

Added directly to word embeddings: $E_{\text{input}} = X_{\text{embed}} + PE$. Allows model to attend by relative positions since PE_{pos+k} can be represented as a linear function of PE_{pos} .

Rotary Position Embedding (RoPE): Applies a 2D rotation matrix to query and key vectors in complex plane:

$$R_{\Theta, m}^d q_m = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 \\ \sin m\theta_1 & \cos m\theta_1 \end{pmatrix} \begin{pmatrix} q_{m,1} \\ q_{m,2} \end{pmatrix}$$

Inner product $(R_{\Theta, m} q_m)^T (R_{\Theta, n} k_n) = q_m^T R_{\Theta, n-m} k_n$ depends purely on relative distance $n - m$.

Part (c): Mathematical Justification of Scaling Factor $\sqrt{d_k}$

Assume components of query $q \in \mathbb{R}^{d_k}$ and key $k \in \mathbb{R}^{d_k}$ are independent random variables with zero mean ($\mathbb{E}[q_i] = \mathbb{E}[k_i] = 0$) and unit variance ($\text{Var}(q_i) = \text{Var}(k_i) = 1$).

Consider the dot product scalar $S = q \cdot k = \sum_{i=1}^{d_k} q_i k_i$: 1. ****Expected Value:****

$$\mathbb{E}[S] = \mathbb{E}\left[\sum_{i=1}^{d_k} q_i k_i\right] = \sum_{i=1}^{d_k} \mathbb{E}[q_i] \mathbb{E}[k_i] = 0$$

2. ****Variance Calculation:****

$$\text{Var}(S) = \text{Var}\left(\sum_{i=1}^{d_k} q_i k_i\right) = \sum_{i=1}^{d_k} \text{Var}(q_i k_i)$$

For independent zero-mean variables: $\text{Var}(q_i k_i) = \mathbb{E}[q_i^2 k_i^2] - (\mathbb{E}[q_i k_i])^2 = \mathbb{E}[q_i^2] \mathbb{E}[k_i^2] - 0 = (1)(1) = 1$.

$$\text{Var}(S) = \sum_{i=1}^{d_k} 1 = d_k$$

Standard deviation of dot product S is $\sqrt{\text{Var}(S)} = \sqrt{d_k}$.

3. ****Why Scaling is Critical:**** For large projection dimensions d_k (e.g. $d_k = 64$ or 128), unscaled dot products have large variance (d_k), pushing inputs into extreme tails of the softmax function. In these regions, softmax outputs approach 1 or 0, and its derivative vanishes ($\frac{\partial \text{softmax}(z_i)}{\partial z_j} \approx 0$). Dividing by $\sqrt{d_k}$ rescales $\text{Var}\left(\frac{q \cdot k}{\sqrt{d_k}}\right) = \frac{\text{Var}(q \cdot k)}{d_k} = \frac{d_k}{d_k} = 1$, keeping gradients well-behaved.

Solution 3: Large Language Models (LLMs)

Part (a): Emergent In-Context Learning (ICL)

In-Context Learning (ICL) is modeled probabilistically as ****implicit Bayesian inference**** over a latent task variable $\tau \sim P(\tau)$. During pretraining on large text corpora, next-token prediction minimizes:

$$\mathcal{L} = - \sum_t \log P(x_t | x_{<t})$$

Since training documents contain diverse tasks, the marginal likelihood integrates over latent tasks:

$$P(x_1, \dots, x_N) = \int P(\tau) \prod_{t=1}^N P(x_t | x_{<t}, \tau) d\tau$$

When an LLM is presented with prompt tokens $C = (x_1, y_1, x_2, y_2, \dots, x_k)$, evaluating $P(y_k | C, x_k)$ implicitly computes the posterior task distribution $P(\tau | C)$ via Bayes' rule:

$$P(\tau | C) \propto P(\tau) P(C | \tau)$$

The model uses prompt examples C as evidence to locate the task vector τ , allowing it to execute new tasks without parameter updates.

Part (b): Masked Self-Attention & Autoregressive Property

The joint probability of a sequence factorizes autoregressively: $p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t | x_{<t})$. To train the entire sequence in parallel without leaking future tokens $x_{>t}$ into x_t , a ****causal mask matrix**** $M \in \mathbb{R}^{T \times T}$ is added to attention logits before softmax:

$$M_{ij} = \begin{cases} 0 & \text{if } i \geq j \\ -\infty & \text{if } i < j \end{cases}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right) V$$

Since $\exp(-\infty) = 0$, attention weights for future positions $j > i$ are zeroed out, ensuring x_t depends only on $x_{\leq t}$.

Part (c): Factual Correctness vs MLE & Solutions

1. ****Why Maximum Likelihood (MLE) Fails for Factual Correctness:**** MLE minimizes $KL(P_{\text{data}} \| P_\theta) = \mathbb{E}_{P_{\text{data}}}[\log \frac{P_{\text{data}}}{P_\theta}]$. Because KL is ****mode-covering**** (zero-avoiding for P_{data}), the model is penalized heavily for assigning zero probability to data sequences, encouraging it to assign non-zero probability to plausible-sounding hallucinations that match superficial n-gram statistics. MLE optimizes fluency, not truth.

2. ****Solutions:****

- ****Direct Preference Optimization (DPO) / RLHF:**** Optimizes preference loss on pair (y_w, y_l) (winning vs losing response):

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

Penalizes ungrounded hallucinations y_l .

- ****Retrieval-Augmented Generation (RAG):**** Conditions generation on verified external knowledge documents R : $P(y | x, R)$.

Part (d): Catastrophic Forgetting & KL Divergence

Fine-tuning updates parameters from pre-trained state θ_{pre} to θ_{ft} on task distribution P_{target} . From a probabilistic perspective, catastrophic forgetting occurs when probability mass assigned to general knowledge distribution $P_{\text{pre}}(y | x)$ drops significantly:

$$KL(P_{\theta_{\text{pre}}}(y | x) \| P_{\theta_{\text{ft}}}(y | x)) \rightarrow \infty$$

To prevent parameter drift away from pre-trained capabilities, fine-tuning objectives add an explicit KL regularization penalty:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{target}}(\theta) + \beta KL(P_{\theta}(y | x) \| P_{\theta_{\text{ref}}}(y | x))$$

where $P_{\theta_{\text{ref}}}$ is the frozen pre-trained model reference.

Solution 4: Diffusion Models

Part (a): Forward and Reverse Process Mathematical Formulation

1. **Forward (Noising) Process** $q(x_{1:T} | x_0)$: A fixed Markov chain that adds Gaussian noise according to variance schedule $\beta_1, \dots, \beta_T$:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. Using the reparameterization trick repeatedly:

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_{t-1} = \dots = \sqrt{\alpha_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

where $\epsilon \sim \mathcal{N}(0, I)$. Thus the marginal distribution at step t is closed-form:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

2. **Reverse (Denoising) Process** $p_{\theta}(x_{0:T})$: Starts from Gaussian noise $p(x_T) = \mathcal{N}(x_T; 0, I)$ and parameterizes transitions:

$$p_{\theta}(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

When conditioned on x_0 , the true posterior $q(x_{t-1} | x_t, x_0)$ is tractable Gaussian via Bayes' rule:

$$q(x_{t-1} | x_t, x_0) = \frac{q(x_t | x_{t-1}, x_0)q(x_{t-1} | x_0)}{q(x_t | x_0)} = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$$

where:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t, \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$$

Part (b): Reverse Decoder Training & ϵ -Prediction Objective

Direct ELBO training minimizes $KL(q(x_{t-1} | x_t, x_0) || p_\theta(x_{t-1} | x_t))$. Since $x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon}{\sqrt{\bar{\alpha}_t}}$, substituting x_0 into posterior mean $\tilde{\mu}_t$ gives:

$$\tilde{\mu}_t(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right)$$

This motivates parameterizing model mean $\mu_\theta(x_t, t)$ with a neural network $\epsilon_\theta(x_t, t)$ predicting noise ϵ :

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$

Simplifying the weighted KL terms yields the **DDPM Mean Squared Error (MSE) Loss** (Ho et al. 2020):

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t \sim U(1, T), x_0 \sim q(x_0), \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right]$$